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# CS403/503 Programming Languages
## Spring 2025
Assignment #4

1. Write a program in C that uses the `qsort` function to sort an array of real numbers (`double`). Make sure your program works with real numbers that differ very small. Paste below only the comparison function passed to `qsort` as a parameter.

Answer:

```
int compare (const void *a, const void *b) {
    return (*(double *)a - *(double *)b);
}
```

2. Problem #5 of Chapter 9 on Page 413.

Answer:

a) Passed by value
   i. value = 2, list = {1, 3, 5, 7, 9}
   ii. value = 2, list = {1, 3, 5, 7, 9}
   iii. value = 2, list = {1, 3, 5, 7, 9}

b) Passed by reference
   i. value = 1, list = {2, 3, 5, 7, 9}
   ii. value = 1, list = {3, 2, 5, 7, 9}
   iii. value = 2, list = {3, 1, 5, 7, 9}

c) Passed by value-result
   a. value = 1, list = {2, 3, 5, 7, 9}
   b. value = 1, list = {3, 2, 5, 7, 9}
   c. value = 2, list = {3, 1, 5, 7, 9}

3. Programming Exercise #8 of Chapter 9 on Page 415.

Answer:

a4_3.h

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```cpp
#ifndef A4_3_H
#define A4_3_H

template<class Type>
int genericSearch(const Type a[], int n, const Type& key) {
    for (int i = 0; i < n; i++) {
        if (a[i] == key) {
            return i;
        }
    }
    return -1;
}

#endif
```

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a4_3.cpp

```cpp
#include <iostream>
#include <stdio.h>
#include "a4_3.h"

using namespace std;

int main() {
    int intArraySize, targetInt, floatArraySize;
    float targetFloat;

    string choice;
    cout << "Enter the type of array you want to search (int/float): ";
    cin >> choice;

    if (choice == "int") {
        cout << "Enter the size of the array: ";
        cin >> intArraySize;

        int intArray[intArraySize];
        cout << "Enter the elements of the array: ";
        for (int i = 0; i < intArraySize; i++) {
            cin >> intArray[i];
        }

        cout << "Enter the target integer: ";
        cin >> targetInt;

        cout << "The target integer is at index " << genericSearch(intArray, intArraySize, targetInt) << endl;
    }
    else if (choice == "float") {
        cout << "Enter the size of the array: ";
        cin >> floatArraySize;

        float floatArray[floatArraySize];
        cout << "Enter the elements of the array: ";
        for (int i = 0; i < floatArraySize; i++) {
            cin >> floatArray[i];
        }
        cout << "Enter the target float: ";
        cin >> targetFloat;
        cout << "The target float is at index " << genericSearch(floatArray, floatArraySize, targetFloat) << endl;
    }

    return 0;
}
```

4. Problem #3 of Chapter 10 on Page 443 (See Figure 10.9 on page 433 for how to draw a stack).

Answer:

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| | |
|---|---|
| ARI  D | dynamic link |
| | static link |
| | return (to C) |
| ARI C | dynamic link |
| | static link |
| | return (to A) |
| ARI A | parameter (flag = False) |
| | dynamic link |
| | static link |
| | return (to B) |
| ARI B | dynamic link |
| | static link |
| | return (to A) |
| ARI A | parameter (flag = True) |
| | dynamic link |
| | static link |
| | return (to BIGSUB) |
| ARI BIGSUB | dynamic link |
| | static link |
| | return (to caller) |